Type-II Fatty-Acid Synthesis (FAS-II): A Dissociated Bacterial Pathway — A Review

1. Executive Summary

Type-II fatty-acid synthesis (FAS-II) is the bacterial route to *de novo* fatty acids. Its defining feature is architectural: unlike the single multifunctional megasynthase of animals and fungi (type-I FAS), FAS-II is **dissociated** — every catalytic step is carried out by a separate, freely diffusing, mono-functional protein, and the growing acyl chain is passed between them while covalently tethered to a small, acidic acyl carrier protein (ACP) through a 4′-phosphopantetheine (4′-PP) arm (PMID: 15052334; PMID: 26104214). This modularity is not a mere biochemical curiosity: it makes individual steps genetically and pharmacologically addressable and underlies the appeal of FAS-II as an antibacterial target.

The chemistry of the pathway is rigidly conserved and can be stated compactly. Acetyl-CoA carboxylase (ACC, the AccABCD complex) produces malonyl-CoA; FabD transfers the malonyl group to holo-ACP; a KAS-III enzyme (FabH) initiates the first chain by condensing acetyl-CoA with malonyl-ACP; and a fixed four-reaction elongation cycle — decarboxylative condensation (FabB/FabF), NADPH-dependent β-keto reduction (FabG), (3R)-hydroxyacyl dehydration (FabZ/FabA), and enoyl reduction (FabI/FabK/FabL/FabV) — extends the chain two carbons at a time. A conditional FabA/FabB branch introduces *cis* double bonds anaerobically without an oxygen-dependent desaturase. The **order** of these reactions is dictated by substrate chemistry and is essentially invariant across bacteria.

What varies, and varies extensively, is **which protein family and cofactor fill each slot**. The clearest illustration is the enoyl-ACP reductase step, performed by at least four non-homologous families (FabI, FabK, FabL, FabV) with divergent cofactor use and inhibitor sensitivity (PMID: 19933806; PMID: 18193820). The unsaturated-fatty-acid branch is similarly implemented by non-orthologous but functionally matched pairs — FabA/FabB in *E. coli*, FabN/FabO in *Enterococcus faecalis* (PMID: 36100979). Finally, the biological importance of the whole system is **conditional**: many Gram-positive pathogens scavenge host fatty acids and can bypass FAS-II inhibition entirely, so essentiality — and therefore drug-target value — is context-dependent (PMID: 34309397; PMID: 38014966). This review synthesizes five confirmed findings into a coherent picture of a pathway that is chemically stereotyped but molecularly plastic.

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2. Definition and Biological Boundaries

What is included

FAS-II is the set of discrete enzymes and carrier proteins that, starting from acetyl-CoA and bicarbonate/ATP, build saturated and *cis*-unsaturated acyl-ACP products of physiologically appropriate chain length (typically C14–C18) that are then handed to membrane-phospholipid biosynthesis. The system comprises:

- **Priming:** acetyl-CoA carboxylase (AccABCD) → malonyl-CoA.
- **Carrier activation and loading:** ACP (holo-AcpP) and FabD (malonyl-CoA:ACP transacylase) → malonyl-ACP.
- **Initiation:** a KAS-III enzyme (FabH) → the first β-ketoacyl-ACP.
- **Iterative elongation:** the four-reaction cycle (condensation → keto-reduction → dehydration → enoyl-reduction).
- **Unsaturation branch:** the FabA/FabB (or equivalent) route to *cis*-unsaturated chains.

What is adjacent but should be treated separately

Several processes are mechanistically related or historically conflated with FAS-II but fall outside its boundary:

- **Type-I FAS (FAS-I).** The animal/fungal megasynthase performs the same chemistry within one (or a few) large multifunctional polypeptides. It is the architectural counterpoint to FAS-II, not a variant of it (PMID: 26104214; PMID: 30248156). Some lineages carry both: *Mycobacterium* uses FAS-I to make medium-chain acyl-CoA primers that FAS-II then **elongates** to very long meromycolate chains (PMID: 26104214; PMID: 11278743).
- **Mitochondrial FAS-II (mtFAS).** Eukaryotic mitochondria retain a bacterial-type, ACP-dependent FAS-II that is essential for respiration and supplies octanoyl-ACP for lipoic-acid synthesis; it is a distinct cellular system that happens to be homologous to bacterial FAS-II (PMID: 20226757; PMID: 19686777).

- **Polyketide synthases (PKS) and non-ribosomal peptide synthetases (NRPS).** These use the same ACP/4′-PP chemistry and share the four-helix-bundle carrier fold, but produce secondary metabolites, not membrane fatty acids (PMID: 37345808; PMID: 28187530).
- **β-Oxidation and exogenous-fatty-acid incorporation.** These are catabolic/salvage routes that intersect FAS-II regulation (via FadR) and can substitute for it physiologically, but they are not part of the synthetic pathway (PMID: 33283913; PMID: 38014966).

Competing definitions

The main definitional tension in the literature is whether FAS-II denotes (a) the strict *de novo* pathway from acetyl-CoA or (b) any dissociated ACP-dependent acyl-chain-elongation system. Mycobacterial FAS-II fits (b) but not (a) — it cannot initiate *de novo* synthesis and only elongates FAS-I products (PMID: 26104214). We treat the dissociated, ACP-dependent architecture as the defining criterion, noting that initiation capacity is lineage-variable.

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3. Mechanistic Overview

The best current model is a strictly ordered, iterative cycle. The order is enforced by substrate chemistry: each enzyme acts only on the product of the previous step.

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Acetyl-CoA + HCO3- + ATP
       │  ACC (AccABCD)  ── biotin carboxylase + carboxyltransferase
       ▼
  Malonyl-CoA
       │  FabD (malonyl-CoA:ACP transacylase)     Holo-AcpP (4'-PP arm)
       ▼                                              ▲ AcpS adds 4'-PP from CoA
  Malonyl-ACP ◄───────────────────────────────────────┘
       │  FabH (KAS-III): acetyl-CoA + malonyl-ACP → C4 β-ketoacyl-ACP + CO2
       ▼
┌───────────────── FOUR-REACTION ELONGATION CYCLE ─────────────────┐
│  (1) Condensation      FabB / FabF (KAS-I/II)                     │
│        acyl-ACP + malonyl-ACP → β-ketoacyl-ACP(+2C) + CO2         │
│  (2) Keto-reduction    FabG  (NADPH)                              │
│        β-ketoacyl-ACP → (3R)-hydroxyacyl-ACP                      │
│  (3) Dehydration       FabZ (all rounds) / FabA (specific)        │
│        (3R)-hydroxyacyl-ACP → trans-2-enoyl-ACP                   │
│  (4) Enoyl-reduction   FabI / FabK / FabL / FabV  (NADH or NADPH) │
│        trans-2-enoyl-ACP → saturated acyl-ACP(+2C)                │
└──────────────────────────────────────────────────────────────────┘
                           │ re-enters cycle (elongate)
                           │
      At C10, trans-2-decenoyl-ACP may branch:
                           ▼
  FabA (isomerase): trans-2-decenoyl-ACP → cis-3-decenoyl-ACP
                           │
                           ▼
  FabB (KAS-I): elongates cis intermediate → cis-unsaturated fatty acids
                           ▼
      Acyl-ACP products (C16:0, C16:1, C18:1 ...) → phospholipid synthesis
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Obligatory steps: priming (ACC), loading (FabD), initiation (a KAS-III or equivalent primer-generating activity), and the four elongation reactions. Without any of these, chain assembly cannot proceed. The condensation → keto-reduction → dehydration → enoyl-reduction order is chemically obligatory: keto-reduction requires the β-keto group that condensation produces; dehydration requires the 3-hydroxy group that keto-reduction produces; enoyl-reduction requires the trans-2 double bond that dehydration produces (PMID: 15052334; PMID: 23322253).

Conditional steps: the FabA/FabB unsaturation branch is engaged only when *cis*-unsaturated chains are needed and only in lineages that use the anaerobic mechanism. It is a diversion from, not a replacement of, the core cycle (PMID: 15052334; PMID: 36515967).

Accessory activities: ACP maturation (AcpS adds 4′-PP; AcpH removes it), branched-chain and hydroxy-fatty-acid tailoring, and regulatory inputs (FadR, FabR) tune output but are not part of the core catalytic cycle (PMID: 37345808; PMID: 33283913; PMID: 32535524).

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4. Major Molecular Players and Active Assemblies

Finding F001 — FAS-II is a dissociated multi-enzyme system distinct from type-I FAS

The foundational fact about FAS-II is architectural. Multiple authoritative reviews establish that in most bacteria fatty acids are made by a group of highly conserved, discrete, mono-functional, freely diffusible proteins — "*Fatty acid biosynthesis is catalyzed in most bacteria by a group of highly conserved proteins known as the type II fatty acid synthase (FAS II) system*" (PMID: 15052334). This is explicitly contrasted with the eukaryotic-type multifunctional FAS-I: the two systems are "*the eukaryotic-like multifunctional enzyme FAS I and the acyl carrier protein (ACP)-dependent FAS II systems, which consists of a series of discrete mono-functional proteins, each catalyzing one reaction in the pathway*" (PMID: 26104214). The dissociated design means each reaction is a separate gene product, and the growing chain is shuttled among them on ACP. Crucially, this same source documents a lineage-specific departure: "*the mycobacterial FAS II is incapable of de novo fatty acid synthesis from acetyl-coenzyme A, but instead elongates medium-chain-length fatty acids previously synthesized by FAS I*" (PMID: 26104214). Thus even the "initiation" boundary of the system is variable.

Finding F002 — Holo-ACP with a 4′-phosphopantetheine arm is the central shuttle

The pathway's logic depends on a single small carrier protein. Apo-ACP is converted to holo-ACP by AcpS-catalyzed transfer of the 4′-PP prosthetic group from coenzyme A; acyl intermediates are then thioesterified to the terminal thiol of that arm. The chain length carried on the arm governs the protein's behaviour and partner recognition: "*The addition of the 4′PP prosthetic group converts apo-AcpP to holo-AcpP (commonly referred to as AcpP). Acylation of the 4′PP prosthetic group gives acyl-AcpP species. The length of the acyl chain determines the properties of the acyl-AcpP*" (PMID: 37345808). ACP is a small acidic four-helix bundle used across FAS, PKS and NRPS systems, and structural studies of freestanding and doublet ACPs confirm the conserved fold and the phosphopantetheinylation-dependent dynamics (PMID:

28187530; PMID: 33416314; PMID: 32128972). Because every catalytic step reads the chain off the same carrier, ACP is simultaneously a substrate presenter and a specificity determinant — the physical hub that makes a "dissociated" pathway function as an integrated system.

Table: Core FAS-II enzymes and their reactions

Step	Reaction	Prototype enzyme(s)	Cofactor	Notes
Priming	acetyl-CoA → malonyl-CoA	ACC (AccABCD)	ATP, biotin, HCO_3^-	Heteromeric in most bacteria; AccC is a drug-target node
Carrier	apo-ACP → holo-ACP	AcpS (adds 4′-PP)	CoA	AcpH reverses
Loading	malonyl-CoA → malonyl-ACP	FabD	—	Malonyl-CoA:ACP transacylase
Initiation	acetyl-CoA + malonyl-ACP → C4 β-ketoacyl-ACP	FabH (KAS-III)	—	Substrate specificity sets branching/chain start
Condensation	acyl-ACP + malonyl-ACP → β-ketoacyl-ACP(+2C)	FabB (KAS-I), FabF (KAS-II)	—	FabB also serves the UFA branch
Keto-reduction	β-ketoacyl-ACP → (3R)-hydroxyacyl-ACP	FabG	NADPH	Highly conserved
Dehydration	(3R)-hydroxyacyl-ACP → trans-2-enoyl-ACP	FabZ (general), FabA (bifunctional)	—	FabA also isomerizes
Enoyl-reduction	trans-2-enoyl-ACP → acyl-ACP(+2C)	FabI, FabK, FabL, FabV	NADH or NADPH	Non-homologous families (F003)
UFA branch	trans-2- → cis-3-decenoyl-ACP; elongation	FabA + FabB (or FabN + FabO)	—	Conditional; anaerobic (F004)

The initiation enzyme (KAS-III/FabH)

FabH condenses an acyl-CoA primer with malonyl-ACP and thereby sets the starting chain and, in some organisms, the branch pattern. The structure of *M. tuberculosis* FabH illustrates how substrate specificity is tuned: mtFabH accepts long-chain acyl-CoA (e.g. C14 myristoyl-CoA) rather than acetyl-CoA because a hydrophobic channel that is blocked by phenylalanine in *E. coli* FabH is opened by a threonine substitution, while a capping helix limits the product to ~C16 (PMID: 11278743). FabH is an actively pursued antibacterial target, with diverse synthetic inhibitor scaffolds reported over 2006–2025 (PMID: 42446656).

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5. Evolutionary and Cell-Biological Variation

Finding F003 — Enoyl-ACP reductase is functionally convergent but family-variable

The most striking example of molecular plasticity is the final elongation step. The same chemical transformation — reduction of the trans-2 double bond — is carried out by **non-homologous** isozyme families with different cofactor preferences and inhibitor sensitivities:

- **FabI** — NADH-dependent, triclosan-sensitive, the *E. coli* prototype and the target of isoniazid, diazaborines and triclosan (PMID: 18193820).
- **FabV** — NADH-dependent but triclosan-**resistant**; present in *Vibrio cholerae* and *Pseudomonas aeruginosa*.
- **FabK** — flavin-dependent and structurally unrelated.
- **FabL** — an additional SDR-family variant.

The physiological consequence of this substitution is dramatic. In *P. aeruginosa* PAO1, "*P. aeruginosa contains two enoyl-ACP reductase isozymes, the previously characterized FabI homologue plus a homologue of FabV, a triclosan-resistant enoyl-ACP reductase recently demonstrated in Vibrio cholerae*" (PMID: 19933806), and deletion of *fabV* rendered the strain "*extremely sensitive to triclosan (>2,000-fold more sensitive than the wild-type strain)*" (PMID: 19933806). This shows that intrinsic triclosan resistance in this organism is due to the identity of the enoyl reductase, not efflux. FabI's status as "*a validated but currently underexploited target for drug discovery*" (PMID: 18193820) is therefore contingent on which reductase a given pathogen actually uses. The reductase steps generally — both FabG and the enoyl reductase — are recognized antimicrobial nodes (PMID: 15726819). In *Sinorhizobium meliloti*, the balance among multiple FabI paralogs (FabI1 vs FabI2) even sets the unsaturated-fatty-acid content of the membrane, with FabI2 being indispensable and only complementable by the *E. faecalis fabI* gene (PMID: 39627703).

Finding F004 — The FabA/FabB anaerobic UFA branch is real but not universal

In *E. coli* and many γ-proteobacteria, *cis*-unsaturation is introduced **without oxygen or a desaturase**: the bifunctional dehydratase/isomerase FabA converts trans-2-decenoyl-ACP to cis-3-decenoyl-ACP, and the KAS-I condensing enzyme FabB preferentially elongates the *cis* intermediate, locking the double bond into the mature chain. This branch, however, is implemented differently across lineages and is not always essential. In *Xanthomonas*, "*the conserved FabA-FabB pathway is considered to be crucial for unsaturated fatty acid (UFA) synthesis in gram-negative bacteria*" yet was shown to be dispensable for UFA synthesis while instead modulating diffusible-signal-factor synthesis (PMID: 36515967). In *Enterococcus faecalis*, the equivalent chemistry is supplied by a non-orthologous but functionally matched pair: "*FabO has been ascribed the same activity as E. coli FabB and we report in vitro evidence that this is the case, whereas FabN is a dehydratase/isomerase, having the activity of E. coli FabA*" (PMID: 36100979). Critically, the reductase must be co-adapted: "*the enoyl-ACP reductase must be matched to the unsaturated fatty acid synthetic genes*" (PMID: 36100979). More broadly, reviews note "*multiple mechanisms to generate unsaturated fatty acids and the accessory components required for branched-chain fatty acid synthesis in Gram-positive bacteria*" (PMID: 15052334). The UFA branch is thus a **conserved outcome achieved by variable, matched enzyme sets** — a textbook case of non-orthologous gene displacement.

Cell-biological and physiological variation

- **Both-systems lineages:** *Mycobacterium* runs FAS-I and FAS-II in series to build mycolic acids; *Streptomyces* has only FAS-II; *Corynebacterium* only FAS-I (PMID: 26104214; PMID: 21204864).
- **Mitochondrial retention:** eukaryotic mitochondria keep a bacterial-type FAS-II essential for respiration and lipoic-acid synthesis (PMID: 20226757; PMID: 19686777).
- **Temperature/physiological state:** UFA content is tuned for membrane fluidity via regulators such as FabR and FadR; in *Shewanella baltica*, cold adaptation is achieved by down-regulating FabR and up-regulating FadR (PMID: 32535524; PMID: 33283913).

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6. Constraints, Dependencies, and Failure Modes

Ordering constraints

The four elongation reactions are locked in sequence by the functional groups they create and consume (see §3). This is why the pathway diagram is effectively a directed cycle with no shortcuts through the core: keto-reduction cannot precede condensation, dehydration cannot precede keto-reduction, and enoyl-reduction cannot precede dehydration. The UFA branch is constrained to a specific chain length — the isomerization operates on the C10 trans-2-decenoyl-ACP intermediate — because that is the intermediate FabA/FabN recognize and FabB/FabO can elongate (PMID: 36100979; PMID: 15052334).

Matching/compatibility constraints

The enoyl reductase and the UFA-synthesis enzymes must be co-adapted; a mismatch breaks UFA production (PMID: 36100979; PMID: 39627703). ACP–partner recognition is chain-length dependent, so acyl intermediates are handed off in an ordered, length-gated fashion rather than promiscuously (PMID: 37345808).

Finding F005 — Essentiality and drug-target value are conditional on host fatty-acid scavenging

FAS-II is the validated target of isoniazid, triclosan, thiolactomycin and diazaborines, and both the ACC/AccC priming node and the FabH/FabI steps are actively pursued (PMID: 26169404; PMID: 18193820). Bioluminescent cellular reporter assays have been built specifically to "*target the enzyme (AccC) catalyzing the biotin carboxylase half-reaction of the acetyl coenzyme A (acetyl-CoA) carboxylase step in the initiation phase of FASII*" (PMID: 26169404). **However, the pathway's essentiality is not absolute.** Many Gram-positive pathogens incorporate exogenous host fatty acids via lipases, fatty-acid kinase systems and auxiliary ACPs, and can thereby bypass FAS-II inhibition. In *Enterococcus faecalis*, the FAS-II pathway and cyclopropane-ring formation were shown to be "*Dispensable during Enterococcus faecalis Systemic Infection*" (PMID: 34309397) because the organism scavenges host fatty acids. The utilization of exogenous unsaturated fatty acids, mediated by bacterial lipases and buffered by human serum albumin, further modulates the impact of FAS-II inhibition (PMID: 38014966). *E. faecalis* even maintains a dedicated auxiliary ACP (EfAcpB) for incorporating exogenous fatty acids alongside the FAS-essential EfAcpA (PMID: 36410271). The practical corollary: **the value of FAS-II as a drug target depends on whether the pathogen, in its infection niche, can obtain fatty acids from the host.**

Failure modes

- **Genetic loss/insufficiency:** loss of an obligatory enzyme is lethal unless salvage is available.
- **Regulatory imbalance:** overproduction of the master regulator FadR increases all acyl-chain yields but at a cost — larger cells, slower growth, intracellular membrane accumulation and abnormal morphology ("*too much FadR is bad*") (PMID: 33283913).
- **Membrane consequences:** perturbing phospholipid composition/fluidity disrupts downstream processes such as MreB rotation (PMID: 33330621); engineering elongation-gene expression (fabH/fabG/fabZ/fabI) reshapes chain-length and UFA output (PMID: 23322253).

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7. Controversies and Open Questions

1. **Where is the true boundary of "FAS-II"?** Mycobacterial FAS-II elongates rather than initiates, forcing a choice between an architecture-based definition (dissociated + ACP-dependent) and a function-based one (*de novo* from acetyl-CoA). We favour the architectural definition but flag that reviews mixing *E. coli* and *Mycobacterium* data can overgeneralize (PMID: 26104214).
2. **How universal is the FabA/FabB anaerobic UFA route?** It is canonical in *E. coli*, dispensable in *Xanthomonas*, and replaced by FabN/FabO in *Enterococcus*. The literature has historically treated it as "the" Gram-negative UFA pathway; recent work shows this is an overstatement (PMID: 36515967; PMID: 36100979).
3. **Is FAS-II a good antibacterial target?** The tension between validated inhibitors (isoniazid, triclosan) and host-fatty-acid bypass is unresolved and organism-specific. Target value must be assessed per pathogen and per infection niche, not asserted for bacteria in general (PMID: 34309397; PMID: 38014966).

4. **Which enoyl reductase, and how redundant?** Organisms carrying multiple non-homologous reductases (e.g. FabI + FabV; multiple FabI paralogs in *Sinorhizobium*) complicate both mechanistic interpretation and inhibitor design (PMID: 19933806; PMID: 39627703).
5. **The dynamics of chain-length gating on ACP.** How ACP's acyl-chain-dependent conformation dictates ordered partner hand-off — and how much of that is programmed versus emergent — remains only partly resolved structurally (PMID: 37345808; PMID: 32128972).
6. **Evolutionary origin.** The dissociated, ACP-dependent design is ancient and conserved (retained even inside mitochondria), whereas the enoyl-reductase family and the UFA-branch enzyme pairs look like later, lineage-specific replacements. FabG (keto-reductase) and the core condensation/carrier machinery are the best candidates for the ancestral, conserved core; the enoyl-reductase slot is the most evolutionarily labile.

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8. Limitations and Knowledge Gaps

- **Organism sampling bias.** Much mechanistic detail derives from *E. coli*; extrapolation to Gram-positives, actinomycetes and environmental bacteria should be cautious given documented non-orthologous displacements.
- **In vitro vs in vivo.** Enzyme kinetics and inhibitor potencies are often measured in isolated systems; in vivo relevance is modulated by redundancy, regulation and host-fatty-acid salvage.
- **Candidate assignments unverified.** The *Pseudomonas putida* (PSEPK) candidate genes named in the working scope (PP_4379, PP_3303, PP_0581, PP_1852) were not experimentally validated in this review; their slot assignments are provisional and rest on homology/annotation rather than functional data.
- **No new primary data.** This review synthesizes existing literature (29 papers); it does not generate new experimental evidence.

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9. Proposed Follow-up Experiments / Actions

1. **Functionally validate the PSEPK candidates.** Test PP_4379 (KAS-III), PP_3303 (KAS-II), PP_0581 (FabG-like reductase) and PP_1852 (NADPH enoyl reductase) by heterologous complementation of *E. coli* conditional mutants (e.g. temperature-sensitive *fabI/fabB/fabH*) and in vitro assays with defined acyl-ACP substrates.
2. **Map enoyl-reductase redundancy per pathogen.** Systematically knock out each reductase family member and measure triclosan/AFN-1252 sensitivity, mirroring the *P. aeruginosa fabV* result, to define target vulnerability organism by organism.
3. **Quantify host-FA bypass.** Under infection-relevant fatty-acid supplementation (with serum albumin), test whether FAS-II inhibitors lose efficacy across a panel of Gram-positive pathogens, extending the *E. faecalis* dispensability finding.
4. **Structural hand-off studies.** Use crosslinked acyl-ACP:enzyme complexes to define chain-length-gated recognition at each step and test whether the UFA branch's C10 specificity can be re-engineered.
5. **Regulatory titration.** Recreate graded FadR/FabR induction to map the quantitative relationship between UFA content, membrane fluidity, cell size and fitness across temperatures.

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10. Key References

- Product diversity and regulation of type II fatty acid synthases. PMID: 15052334
- The Molecular Genetics of Mycolic Acid Biosynthesis. PMID: 26104214
- Fatty acid biosynthesis in actinomycetes. PMID: 21204864
- The acyl carrier proteins of lipid synthesis are busy having other affairs. PMID: 37345808
- Inhibitors of FabI, an enzyme drug target in the bacterial fatty acid biosynthesis pathway. PMID: 18193820
- The reductase steps of the type II fatty acid synthase as antimicrobial targets. PMID: 15726819
- Triclosan resistance of *Pseudomonas aeruginosa* PAO1 is due to FabV. PMID: 19933806
- Unsaturated fatty acid synthesis in *Enterococcus faecalis* requires a specific enoyl-ACP reductase. PMID: 36100979

- The FabA-FabB Pathway Is Not Essential for UFA Synthesis but Modulates DSF Synthesis (*Xanthomonas*). PMID: 36515967
- Type II Fatty Acid Synthesis Pathway and Cyclopropane Ring Formation Are Dispensable during *Enterococcus faecalis* Systemic Infection. PMID: 34309397
- Elucidating the impact of bacterial lipases, human serum albumin, and FASII inhibition on exogenous fatty-acid utilization. PMID: 38014966
- Discovery of bacterial FAS-II inhibitors using a cellular bioluminescent reporter assay (AccC). PMID: 26169404
- Crystal structure of the *M. tuberculosis* β-ketoacyl-ACP synthase III (FabH). PMID: 11278743
- FabH — Promising Novel Antibacterial Target and Its Inhibitors. PMID: 42446656
- The low ENR activity of FabI2 drives high UFA composition in *Sinorhizobium meliloti*. PMID: 39627703
- The *E. coli* FadR transcription factor: Too much of a good thing? PMID: 33283913
- Transcription factors FabR and FadR regulate cold adaptability of *Shewanella baltica*. PMID: 32535524
- Mitochondrial fatty acid synthesis and respiration. PMID: 20226757
- Mitochondrial fatty acid synthesis — an adopted set of enzymes. PMID: 19686777
- Correlations between FAS elongation-cycle gene expression and fatty-acid production in *E. coli*. PMID: 23322253
- Structural study of *E. faecalis* ACP and its interaction with FAS enzymes. PMID: 36410271
- Expression of a recombinant, 4′-phosphopantetheinylated, active *M. tuberculosis* FAS I in *E. coli*. PMID: 30248156
- Alteration of membrane fluidity/phospholipid composition perturbs MreB rotation. PMID: 33330621
- Structural/dynamic characterization of a freestanding ACP (friulimicin biosynthesis). PMID: 28187530
- Solution structure of a doublet ACP from prodigiosin biosynthesis. PMID: 33416314
- Conformational flexibility of CoA and its impact on 4′-PP transfer to ACP. PMID: 32128972

Prepared as a commissioned review synthesis on Type-II fatty-acid synthesis. Evidence base: 29 papers; 5 confirmed findings. The chemistry and step-order of FAS-II are strongly conserved; the enzyme families and cofactors filling each catalytic slot, and the pathway's physiological essentiality, are lineage- and niche-dependent.
